An Expedient Three-Component Synthesis of Tertiary Benzylamines

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Abstract: A Mannich-like zinc-mediated three-component reaction of aromatic halides, amines, and paraformaldehyde is described. This procedure, which involves the in situ formation of arylzinc reagents, allows the straightforward synthesis of a range of functionalized tertiary benzylamines.

Key words: multicomponent reactions, amines, catalysis, zinc, organometallic reagents

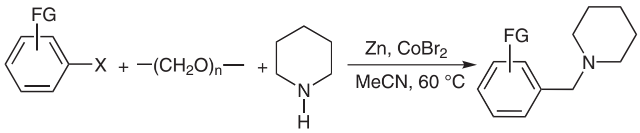
Benzylamines are compounds of high interest that have potential application both as reliable precursors of fine chemical derivatives[1] and as pharmacologically active compounds.[2] These attractive properties have made them prominent synthetic targets for organic chemists. Several general strategies can be envisaged for the preparation of the benzylamine moiety, including reductive amination of aldehydes,[3] hydroamination of alkenes,[4] nitrile reduction,[5] amination of alcohols,[6] nucleophilic substitution of halides by amines,[7] Suzuki–Miyaura coupling with trifluoroborates,[8] and proton shift of triazenes.[9]

Curiously, despite their high synthetic interest, multicomponent procedures related to the Mannich reaction have not been employed so far in a systematic study dedicated to the formation of functionalized tertiary benzylamines.[10] Such strategies, however, would appear to be very reliable due to cost-effective starting compounds and the opportunity to prepare easily libraries of compounds.

Over the last few years, our group has developed a Mannich-type three-component reaction between organozinc reagents, amines, and aldehyde derivatives.[11] In a very recent study, we disclosed the possibility of operating under Barbier-like conditions, starting from organic halides as organozinc precursors.[12] Herein, we confirm the possible in situ metalation of aryl bromides with zinc dust by applying this strategy to the preparation of a range of functionalized tertiary benzylamines.

In the first series of experiments we investigated the scope of the aryl halide in three-component couplings with piperidine as model amine and paraformaldehyde as a formaldehyde synthetic equivalent. The most general and convenient reaction conditions were defined as follows: acetonitrile was used as solvent, zinc dust as the reductive metal, and cobalt(II) bromide as the catalyst[13] with the aromatic halide (3 equiv[14]), amine (1 equiv), and paraformaldehyde (2 equiv) allowed to react at 60 °C; under these conditions, reactions generally appeared to be complete in one hour.[15,16] Results are reported in Table 1.

Table 1 Coupling between Aryl Halides, Paraformaldehyde, and Piperidine[a]



1a–i

Entry	X	FG	Time (h)	Product	Yield (%)
1	Cl	H	1	**1a**	–
2	Br	H	1	**1a**	96
3	I	H	0.5	**1a**	70[b]
4	Br	4-*i*-Pr	1	**1b**	87
5	Br	4-OMe	1	**1c**	87
6	Br	3-CF_3	1	**1d**	80
7	Br	4-CO_2Et	1	**1e**	70
8	Br	3-CO_2Et	1	**1f**	91
9	Br	3-F	1	**1g**	87
10	Br	3-Cl	1	**1h**	83
11	Br	2-Cl	1	**1i**	–[c]
12	I	2-Cl	2.5	**1i**	76[b]

[a] Reagents: Zn dust (6 g), $CoBr_2$ (0.66 g), ArBr (30 mmol), paraformaldehyde (20 mmol), piperidine (10 mmol), MeCN (40 mL).
[b] Caution, vigorous reaction: conducted at r.t.
[c] Principally observed as a transient species.

The first point to note is that while chlorobenzene is inactive in the reaction (Table 1, entry 1), bromobenzene is a very reliable halide, providing the corresponding three-component coupling product **1a** in almost quantitative yield after one hour at 60 °C (Table 1, entry 2). Iodobenzene also undergoes the reaction (Table 1, entry 3). In this case, heating is not necessary and actually not recommended because a sudden exothermic process tends to develop upon addition of cobalt(II) bromide in the reaction medium. Accordingly, in most reactions, functionalized bromobenzenes possess a more appropriate reactivity and

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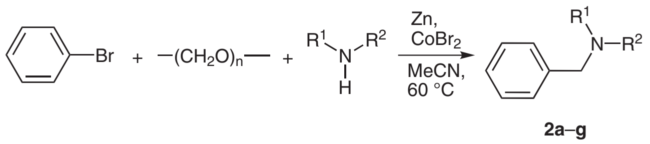
also constitute more cost-effective starting compounds (Table 1, entries 4–10). However, the use of a functionalized iodobenzene might constitute a suitable alternative to the corresponding bromobenzene for reactions in which the latter would not undergo the coupling. This assertion is exemplified by the comparison of Table 1, entries 11 and 12. Indeed, if 1-bromo-2-chlorobenzene is used as the starting aryl bromide, a mixture of products is observed (Table 1, entry 11), whereas a very satisfactory result is obtained by using 1-chloro-2-iodobenzene as the organozinc precursor (Table 1, entry 12).

In a second series of experiments we examined the scope of the secondary amine. To this end, bromobenzene was chosen as the model halide. The experimental conditions were typically identical to those already described above. Results are displayed in Table 2.

Table 2 Coupling between Bromobenzene, Paraformaldehyde, and Secondary Amines[a]



Entry	Amine R^1R^2NH	Time (h)	Product	Yield (%)
1	pyrrolidine	2	**2a**	97
2	tetrahydroisoquinoline	2	**2b**	84
3	morpholine	2	**2c**	70
4	*N*-phenylpiperazine	1.5	**2d**	89
5	indoline	0.5	**2e**	71
6	*N*-methylaniline	1	**2f**	97
7	diethylamine	2	**2g**	64

[a] Reagents: Zn dust (6 g), $CoBr_2$ (0.66 g), PhBr (30 mmol), paraformaldehyde (20 mmol), amine (10 mmol), MeCN (40 mL).

A large range of secondary amines are usable in the process. However, it should be noted that diethylamine led to the corresponding coupling product in only moderate yield (Table 2, entry 7). This limited yield might be simply attributed to a partial loss of the starting amine upon heating, due to its high volatility.

In a third set of experiments we attempted to verify the versatility of the reaction by realizing 'cross-couplings' between paraformaldehyde and a range of bromobenzenes and amines. We additionally tried to expand further the scope of the procedure by checking the behavior of other organic halides and particularly heteroaromatic halides in the reaction. This could lead to the straightforward formation of some benzylamines analogues. Miscellaneous results are reported in Table 3.

The first point to note is that, apart from the possibility to operate starting from miscellaneous amines and bromobenzenes (Table 3, entries 1–4), an aromatic bromide derived from naphthalene was efficient in the process (Table 3, entry 5). Halopyridines did not undergo the three-component reaction (Table 3, entries 6–8) whereas halothiophenes were very efficient in the process, as attested by the successful coupling of 2- and 3-bromothiophene with amines and paraformaldehyde (Table 3, entries 9 and 10).

In conclusion, the results reported in this study highlight the versatility and the reliability of this novel three-component reaction between aryl halides, amines, and paraformaldehyde for the synthesis of benzylamines. Indeed, we show that a large range of halobenzenes and amines can be employed in the process to furnish functionalized tertiary benzylamines and analogues with short reaction times and high yields. Further extensions of this process, in particular the generalization of the use of Barbier-like conditions starting from aryl bromides, are currently in progress.

MeCN (analytical grade) and reagents were purchased from commercial suppliers and used without further purification. All reactions were monitored by gas chromatography (GC) using a Varian 3400 chromatograph equipped with a 5 m SGE BP1 column. Melting points were determined on an Electrothermal IA 9100 capillary melting apparatus and are uncorrected. IR spectra were performed on a Bruker Tensor 27 ATR-FTIR spectrophotometer. NMR spectra were recorded at 400 MHz (^{1}H), 100 MHz (^{13}C), and 376 MHz (^{19}F) on a Bruker Avance II 400 spectrometer in $CDCl_3$ relative to the residual $CHCl_3$ signal. Mass spectra were measured on a Finnigan GC/MS GCQ Thermoquest spectrometer. Elemental analyses were performed at the 'Service de Microanalyse', CNRS, Gif-sur-Yvette (France).

Tertiary Benzylamines 1–3; General Procedure

A dried 100-mL round-bottomed flask was flushed with argon and charged with MeCN (40 mL). Zn dust (6 g, 92 mmol), TFA (306 mg, 200 μL, 2.68 mmol), and 1,2-dibromoethane (872 mg, 400 μL, 4.64 mmol) were added to the soln which was heated with stirring until production of gas. Heating was then stopped and the soln stirred at r.t. for an additional 10 min. The aryl bromide (30 mmol), paraformaldehyde (0.6 g, ~20 mmol), amine (10 mmol), and $CoBr_2$ (0.66 g, 3 mmol) were added to the soln, which was heated at 60 °C until completion of the reaction (typically 1–2 h). The mixture was poured into sat. aq NH_4Cl soln (150 mL) and extracted with CH_2Cl_2 (2 × 100 mL). The combined organic fractions were concentrated to dryness and the crude oil was taken up with Et_2O (150 mL). After complete dissolution, concd H_2SO_4 (0.5–0.75 mL) was added carefully to the vigorously stirred soln. After stirring for 5 min, the resulting ammonium salt was filtered and washed with Et_2O (2 × 50 mL). The solid was then poured under stirring into 5% aq NaOH soln (100 mL) and extracted with CH_2Cl_2 (2 × 100 mL). The combined organic fractions were dried (Na_2SO_4) and concentrated to dryness to yield the analytically pure (>97% GC) benzylamine.

1-Benzylpiperidine (1a)[8]

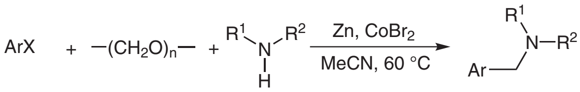
Light yellow oil; yield: 1.69 g (96%).

^{1}H NMR: δ = 7.32–7.11 (m, 5 H), 3.42 (s, 2 H), 2.32 (br s, 4 H), 1.59–1.44 (m, 4 H), 1.41–1.28 (m, 2 H).

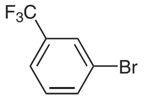
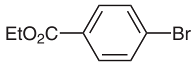
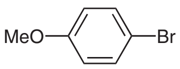
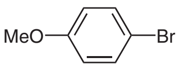
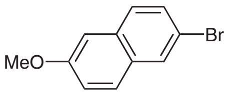
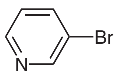
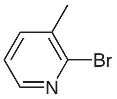
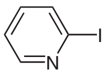
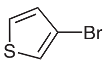
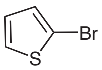
^{13}C NMR: δ = 138.2, 129.3, 128.1, 127.0, 63.8, 54.4, 25.9, 24.3.

MS: *m/z* (%) = 177 (24), 176 (43), 175 (45), 174 (100), 98 (26), 92 (8), 91 (76), 84 (32), 70 (6), 65 (9).

Table 3 Coupling between Aryl Halides, Paraformaldehyde, and Amines[a]



3a–g

Entry	ArX	Amine R^1R^2NH	Time (h)	Product	Yield (%)
1		morpholine	1	**3a**	75
2		diethylamine	2	**3b**	83
3		morpholine	1	**3c**	75
4		*N*-phenylpiperazine	1	**3d**	55
5		piperidine	1.5	**3e**	66
6[b]		tetrahydroisoquinoline	2	–	–
7[b]		piperidine	1.5	–	–
8[b]		piperidine	2	–	–[c]
9		piperidine	1	**3f**	72
10		tetrahydroisoquinoline	1	**3g**	85

[a] Reagents: Zn dust (6 g), $CoBr_2$ (0.66 g), ArX (30 mmol), paraformaldehyde (20 mmol), amine (10 mmol), MeCN (40 mL).
[b] Conducted on a 5 mmol scale using half of the amounts stated above.
[c] 2,2′-Bipyridine was the major product of the reaction.

1-(4-Isopropylbenzyl)piperidine (1b)
Colorless oil; yield: 1.89 g (87%).

^{1}H NMR: δ = 7.16 (d, *J* = 8.1 Hz, 2 H), 7.09 (d, *J* = 8.1 Hz, 2 H), 3.37 (s, 2 H), 2.81 (hept, *J* = 6.9 Hz, 1 H), 2.30 (br s, 4 H), 1.54–1.44 (m, 4 H), 1.40–1.29 (m, 2 H), 1.17 (d, *J* = 6.9 Hz, 6 H).

^{13}C NMR: δ = 147.5, 135.8, 129.3, 126.1, 63.6, 54.5, 33.8, 25.9, 24.4, 24.1.

MS: *m*/*z* (%) = 218 (20), 217 (56), 216 (100), 174 (11), 134 (8), 133 (66), 118 (5), 117 (17), 115 (10), 105 (24), 98 (14), 92 (9), 91 (19), 84 (55), 56 (7).

Anal. Calcd for $C_{15}H_{23}N$: C, 82.89; H, 10.67; N, 6.44. Found: C, 82.54; H, 10.63; N, 6.35.

1-(4-Methoxybenzyl)piperidine (1c)[8]
Yellow oil; yield: 1.79 g (87%).

^{1}H NMR: δ = 7.24 (d, *J* = 8.7 Hz, 2 H), 6.86 (d, *J* = 8.7 Hz, 2 H), 3.81 (s, 3 H), 3.44 (s, 2 H), 2.38 (br s, 4 H), 1.64–1.52 (m, 4 H), 1.49–1.35 (m, 2 H).

^{13}C NMR: δ = 157.2, 129.1, 128.9, 112.1, 61.8, 53.9, 52.9, 24.5, 23.0.

MS: *m*/*z* (%) = 206 (13), 205 (51), 204 (61), 123 (5), 122 (32), 121 (100), 98 (7), 91 (10), 85 (7), 84 (29), 78 (5), 77 (9).

1-[3-(Trifluoromethyl)benzyl]piperidine (1d)[15]
Light-yellow oil; yield: 1.95 g (80%).

^{1}H NMR: δ = 7.49–7.43 (m, 1 H), 7.34–7.30 (m, 2 H), 7.28 (t, *J* = 7.7 Hz, 1 H), 3.38 (s, 2 H), 2.25 (br s, 4 H), 1.55–1.40 (m, 4 H), 1.38–1.25 (m, 2 H).

^{13}C NMR: δ = 140.0, 132.3, 130.5 (q, *J* = 32.0 Hz), 128.5, 125.6 (q, *J* = 3.8 Hz), 124.3 (q, *J* = 272.2 Hz), 123.6 (q, *J* = 3.8 Hz), 63.3, 54.5, 25.9, 24.3.

^{19}F NMR: δ = –62.48.

MS: *m*/*z* (%) = 244 (5), 243 (32), 242 (100), 200 (5), 159 (42), 109 (8), 98 (13).

Ethyl 4-[(Piperidin-1-yl)methyl]benzoate (1e)
Light-orange oil; yield: 1.73 g (70%).

^{1}H NMR: δ = 7.92 (d, J = 8.3 Hz, 2 H), 7.35 (d, J = 8.3 Hz, 2 H), 4.30 (q, J = 7.1 Hz, 2 H), 3.50 (s, 2 H), 2.35 (br s, 4 H), 1.61–1.49 (m, 4 H), 1.44–1.23 (m, 5 H).

^{13}C NMR: δ = 166.6, 129.5, 129.4, 129.2, 63.1, 60.9, 54.4, 25.5, 24.0, 14.3.

MS: *m/z* (%) = 248 (7), 247 (47), 246 (100), 218 (21), 202 (10), 163 (46), 135 (13), 107 (8), 98 (15), 91 (6), 84 (24).

Anal. Calcd for $C_{15}H_{21}NO_2$: C, 72.84; H, 8.56; N, 5.66. Found: C, 72.76; H, 8.39; N, 5.59.

Ethyl 3-[(Piperidin-1-yl)methyl]benzoate (1f)
Yellow oil; yield: 2.25 g (91%).

^{1}H NMR: δ = 7.95–7.81 (m, 2 H), 7.53 (d, J = 7.5 Hz, 1 H), 7.33 (t, J = 7.6 Hz, 1 H), 4.31 (q, J = 7.1 Hz, 2 H), 3.49 (s, 2 H), 2.35 (br s, 4 H), 1.62–1.48 (m, 4 H), 1.43–1.27 (m, 5 H).

^{13}C NMR: δ = 166.7, 133.9, 130.4, 130.4, 128.4, 128.3, 63.2, 61.0, 54.3, 25.7, 24.1, 14.4.

MS: *m/z* (%) = 248 (7), 247 (43), 246 (100), 218 (18), 163 (33), 135 (5), 119 (19), 98 (13), 91 (8), 84 (22).

Anal. Calcd for $C_{15}H_{21}NO_2$: C, 72.84; H, 8.56; N, 5.66. Found: C, 72.77; H, 8.68; N, 5.71.

1-(3-Fluorobenzyl)piperidine (1g)[16]
Light-yellow oil; yield: 1.68 g (87%).

^{1}H NMR: δ = 7.23–7.14 (m, 1 H), 7.05–6.96 (m, 2 H), 6.90–6.81 (m, 1 H), 3.41 (s, 2 H), 2.32 (br s, 4 H), 1.61–1.46 (m, 4 H), 1.42–1.26 (m, 2 H).

^{13}C NMR: δ = 162.9 (d, J = 245.2 Hz), 141.1, 129.5 (d, J = 8.2 Hz), 124.7 (d, J = 2.8 Hz), 115.9 (d, J = 21.2 Hz), 113.8 (d, J = 21.1 Hz), 63.1 (d, J = 1.8 Hz), 54.4, 25.8, 24.2.

^{19}F NMR: δ = –113.92.

MS: *m/z* (%) = 194 (5), 193 (40), 192 (100), 109 (10), 98 (15), 84 (58).

1-(3-Chlorobenzyl)piperidine (1h)[16]
Yellow oil; yield: 1.74 g (83%).

^{1}H NMR: δ = 7.26 (s, 1 H), 7.21–7.06 (m, 3 H), 3.36 (s, 2 H), 2.29 (br s, 4 H), 1.60–1.43 (m, 4 H), 1.41–1.23 (m, 2 H).

^{13}C NMR: δ = 140.9, 134.0, 129.4, 129.1, 127.2, 127.0, 63.2, 54.5, 25.9, 24.3.

MS: *m/z* (%) = 212 (14), 211 (20), 210 (80), 209 (43), 208 (100), 127 (17), 126 (9), 125 (45), 99 (6), 98 (23), 91 (8), 89 (14), 84 (60), 70 (7), 56 (8).

1-(2-Chlorobenzyl)piperidine (1i)[16]
Light-yellow oil; yield: 1.59 g (76%).

^{1}H NMR: δ = 7.42 (dd, J = 7.7, 1.6 Hz, 1 H), 7.25 (dd, J = 7.9, 1.4 Hz, 1 H), 7.20–7.03 (m, 2 H), 3.50 (s, 2 H), 2.37 (m, 4 H), 1.58–1.26 (m, 6 H).

^{13}C NMR: δ = 136.5, 134.2, 130.6, 129.3, 127.8, 126.5, 59.9, 54.7, 26.1, 24.3.

MS: *m/z* (%) = 212 (5), 211 (14), 210 (53), 209 (40), 208 (100), 174 (6), 127 (10), 125 (34), 98 (14), 89 (6).

1-Benzylpyrrolidine (2a)[6e]
Light-yellow oil; yield: 1.57 g (97%).

^{1}H NMR: δ = 7.32–7.11 (m, 5 H), 3.55 (s, 2 H), 2.49–2.38 (m, 4 H), 1.78–1.62 (m, 4 H).

^{13}C NMR: δ = 139.2, 128.9, 128.2, 126.9, 60.7, 54.2, 23.4.

MS: *m/z* (%) = 162 (16), 161 (29), 160 (84), 132 (8), 118 (6), 92 (10), 91 (100), 84 (24), 70 (70), 65 (17).

2-Benzyl-1,2,3,4-tetrahydroisoquinoline (2b)[17]
Light-brown oil; yield: 1.88 g (84%).

^{1}H NMR: δ = 7.48–7.25 (m, 5 H), 7.19–7.09 (m, 3 H), 7.05–6.94 (m, 1 H), 3.73 (s, 2 H), 3.67 (s, 2 H), 2.93 (t, J = 5.9 Hz, 2 H), 2.79 (t, J = 5.9 Hz, 2 H).

^{13}C NMR: δ = 138.2, 134.8, 134.3, 129.2, 128.7, 128.3, 127.2, 126.6, 126.1, 125.6, 62.7, 56.0, 50.6, 29.1.

MS: *m/z* (%) = 224 (5), 223 (44), 222 (100), 146 (5), 133 (5), 132 (34), 130 (7), 104 (6), 92 (5), 91 (46), 78 (6).

4-Benzylmorpholine (2c)[6e]
Yellow oil; yield: 1.24 g (70%).

^{1}H NMR: δ = 7.40–7.19 (m, 5 H), 3.79–3.66 (m, 4 H), 3.52 (s, 2 H), 2.54–2.36 (m, 4 H).

^{13}C NMR: δ = 137.6, 129.2, 128.3, 127.2, 67.0, 63.5, 53.6.

MS: *m/z* (%) = 178 (19), 177 (58), 176 (26), 147 (7), 146 (59), 118 (7), 105 (7), 104 (9), 100 (6), 92 (11), 91 (100), 86 (75), 65 (14), 56 (6).

1-Benzyl-4-phenylpiperazine (2d)[18]
Light-brown solid; yield: 2.25 g (89%); mp 52–54 °C.

^{1}H NMR: δ = 7.54–7.29 (m, 7 H), 7.03 (d, J = 7.9 Hz, 2 H), 6.96 (t, J = 7.3 Hz, 1 H), 3.67 (s, 2 H), 3.37–3.22 (m, 4 H), 2.78–2.63 (m, 4 H).

^{13}C NMR: δ = 151.5, 138.1, 129.3, 129.2, 128.4, 127.3, 119.7, 116.1, 63.2, 53.2, 49.2.

MS: *m/z* (%) = 253 (19), 252 (100), 251 (12), 146 (14), 145 (9), 119 (35), 118 (27), 104 (16), 103 (8), 102 (11), 91 (50).

1-Benzyl-2,3-dihydro-1*H*-indole (2e)[19]
Brown oil; yield: 1.49 g (71%).

^{1}H NMR: δ = 7.65–7.43 (m, 5 H), 7.37–7.22 (m, 2 H), 6.91 (t, J = 7.4 Hz, 1 H), 6.73 (d, J = 7.8 Hz, 1 H), 4.45 (s, 2 H), 3.51 (t, J = 8.3 Hz, 2 H), 3.17 (t, J = 8.3 Hz, 2 H).

^{13}C NMR: δ = 152.8, 138.7, 130.2, 128.7, 128.1, 127.6, 127.3, 124.7, 118.0, 107.3, 53.9, 53.8, 28.8.

MS: *m/z* (%) = 210 (18), 209 (100), 208 (22), 132 (24), 118 (37), 117 (10), 92 (7), 91 (47), 65 (8).

***N*-Benzyl-*N*-methylaniline (2f)**[20]
Orange oil; yield: 1.92 g (97%).

^{1}H NMR: δ = 7.65–7.46 (m, 7 H), 7.11–6.95 (m, 3 H), 4.79 (s, 2 H), 3.27 (s, 3 H).

^{13}C NMR: δ = 150.0, 139.3, 129.5, 128.9, 127.2, 127.1, 116.9, 112.7, 56.9, 38.8.

MS: *m/z* (%) = 198 (14), 197 (100), 196 (44), 180 (5), 120 (44), 118 (5), 106 (9), 104 (6), 91 (44), 77 (10), 65 (9).

***N*-Benzyldiethylamine (2g)**[6e]
Light-yellow oil; yield: 1.05 g (64%).

^{1}H NMR: δ = 7.34–7.11 (m, 5 H), 3.50 (s, 2 H), 2.46 (q, J = 7.1 Hz, 4 H), 0.98 (t, J = 7.1 Hz, 6 H).

^{13}C NMR: δ = 139.6, 129.0, 128.1, 126.7, 57.5, 46.7, 11.6.

MS: *m/z* (%) = 163 (10), 162 (6), 149 (6), 148 (59), 92 (8), 91 (100), 65 (7).

4-[3-(Trifluoromethyl)benzyl]morpholine (3a)[15]
Light-yellow oil; yield: 1.84 g (75%).

^{1}H NMR: δ = 7.53 (br s, 1 H), 7.49–7.31 (m, 3 H), 3.70–3.58 (m, 4 H), 3.47 (s, 2 H), 2.45–2.28 (m, 4 H).

^{13}C NMR: δ = 138.9, 132.4 (d, *J* = 1.0 Hz), 130.6 (q, *J* = 32.1 Hz), 128.7, 124.2 (q, *J* = 272.3 Hz), 125.7 (q, *J* = 3.8 Hz), 124.1 (q, *J* = 3.8 Hz), 66.9, 62.8, 53.6.

^{19}F NMR: δ = –62.61.

MS: *m*/*z* (%) = 246 (30), 245 (99), 244 (63), 215 (13), 214 (100), 200 (5), 186 (9), 172 (11), 160 (19), 159 (92), 141 (12), 110 (6), 109 (18), 100 (12), 86 (45).

Ethyl 4-[(Diethylamino)methyl]benzoate (3b)
Orange oil; yield: 1.96 g (83%).

^{1}H NMR: δ = 7.99 (d, *J* = 8.5 Hz, 2 H), 7.42 (d, *J* = 8.5 Hz, 2 H), 4.37 (q, *J* = 7.1 Hz, 2 H), 3.61 (s, 2 H), 2.52 (q, *J* = 7.1 Hz, 4 H), 1.39 (t, *J* = 7.1 Hz, 3 H), 1.04 (t, *J* = 7.1 Hz, 6 H).

^{13}C NMR: δ = 166.7, 145.5, 129.4, 129.0, 128.6, 60.8, 57.4, 46.9, 14.3, 11.7.

MS: *m*/*z* (%) = 235 (9), 221 (12), 220 (100), 164 (8), 163 (76), 135 (12), 107 (7).

Anal. Calcd for $C_{14}H_{21}NO_2$: C, 71.46; H, 8.99; N, 5.95. Found: C, 71.15; H, 8.83; N, 5.67.

4-(4-Methoxybenzyl)morpholine (3c)[8]
Yellow oil; yield: 1.56 g (75%).

^{1}H NMR: δ = 7.15 (d, *J* = 8.2 Hz, 2 H), 6.78 (d, *J* = 8.2 Hz, 2 H), 3.72 (s, 3 H), 3.67–3.55 (m, 4 H), 3.36 (s, 2 H), 2.34 (br s, 4 H).

^{13}C NMR: δ = 158.8, 130.4, 129.7, 113.6, 67.0, 62.8, 55.2, 53.5.

MS: *m*/*z* (%) = 208 (7), 207 (51), 206 (17), 176 (19), 135 (5), 134 (10), 122 (20), 121 (100), 91 (10), 86 (33), 77 (11).

1-(4-Methoxybenzyl)-4-phenylpiperazine (3d)[3b]
Light-brown solid; yield: 1.57 g (55%); mp 105–107 °C.

^{1}H NMR: δ = 7.35–7.20 (m, 4 H), 7.00–6.81 (m, 5 H), 3.82 (s, 3 H), 3.55 (s, 2 H), 3.30–3.13 (s, 4 H) 2.71–2.55 (s, 4 H).

^{13}C NMR: δ = 158.8, 151.3, 130.5, 129.1, 119.7, 116.0, 113.7, 62.4, 55.3, 52.9, 49.1.

MS: *m*/*z* (%) = 283 (22), 282 (100), 281 (7), 254 (15), 176 (9), 175 (9), 162 (7), 161 (12), 150 (5), 149 (68), 148 (44), 134 (15), 133 (7), 132 (10), 121 (45), 118 (6), 106 (10), 104 (8), 91 (9), 77 (9), 56 (9).

1-[(6-Methoxynaphthalen-2-yl)methyl]piperidine (3e)
Light-brown solid; yield: 1.68 g (66%); mp 64–66 °C.

^{1}H NMR: δ = 7.69–7.54 (m, 3 H), 7.39 (d, *J* = 8.4 Hz, 1 H), 7.09–7.01 (m, 2 H), 3.84 (s, 3 H), 3.55 (s, 2 H), 2.36 (br s, 4 H), 1.60–1.47 (m, 4 H), 1.42–1.29 (m, 2 H).

^{13}C NMR: δ = 157.5, 133.8, 129.2, 128.7, 128.3, 127.8, 126.6, 118.7, 105.6, 63.8, 55.3, 54.5, 25.8, 24.3.

MS: *m*/*z* (%) = 256 (6), 255 (28), 254 (22), 173 (9), 172 (77), 171 (100), 157 (12), 156(8), 129 (11), 128 (34), 84 (51).

Anal. Calcd for $C_{17}H_{21}NO$: C, 79.96; H, 8.29; N, 5.49. Found: C, 79.78; H, 8.31; N, 5.39.

1-[(Thiophen-3-yl)methyl]piperidine (3f)
Yellow oil; yield: 1.31 g (72%).

^{1}H NMR: δ = 7.31–7.22 (m, 1 H), 7.14–7.09 (m, 1 H), 7.06 (d, *J* = 5.9 Hz, 1 H), 3.52 (s, 2 H), 2.39 (br s, 4 H), 1.67–1.52 (m, 4 H), 1.49–1.34 (m, 2 H).

^{13}C NMR: δ = 139.3, 128.8, 125.1, 122.7, 58.3, 54.4, 25.9, 24.3.

MS: *m*/*z* (%) = 182 (37), 181 (14), 180 (44), 98 (15), 97 (80), 85 (6), 84 (100), 56 (9), 53 (6).

Anal. Calcd for $C_{10}H_{15}NS$: C, 66.25; H, 8.34; N, 7.73. Found: C, 65.91; H, 8.17; N, 7.68.

2-[(Thiophen-2-yl)methyl]-1,2,3,4-tetrahydroisoquinoline (3g)[3c]
Orange semi-solid oil; yield: 1.96 g (85%).

^{1}H NMR: δ = 7.35 (dd, *J* = 4.7, 1.6 Hz, 1 H), 7.29–7.17 (m, 3 H), 7.13–7.02 (m, 3 H), 4.00 (s, 2 H), 3.81 (s, 2 H), 3.02 (t, *J* = 5.9 Hz, 2 H), 2.89 (t, *J* = 5.9 Hz, 2 H).

^{13}C NMR: δ = 142.0, 134.8, 134.4, 128.8, 126.8, 126.6, 126.3, 126.1, 125.8, 125.2, 57.0, 55.9, 50.4, 29.3.

MS: *m*/*z* (%) = 229 (8), 228 (27), 145 (18), 133 (9), 132 (100), 117 (9), 105 (6), 104 (8), 103 (5), 97 (47), 78 (6).

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